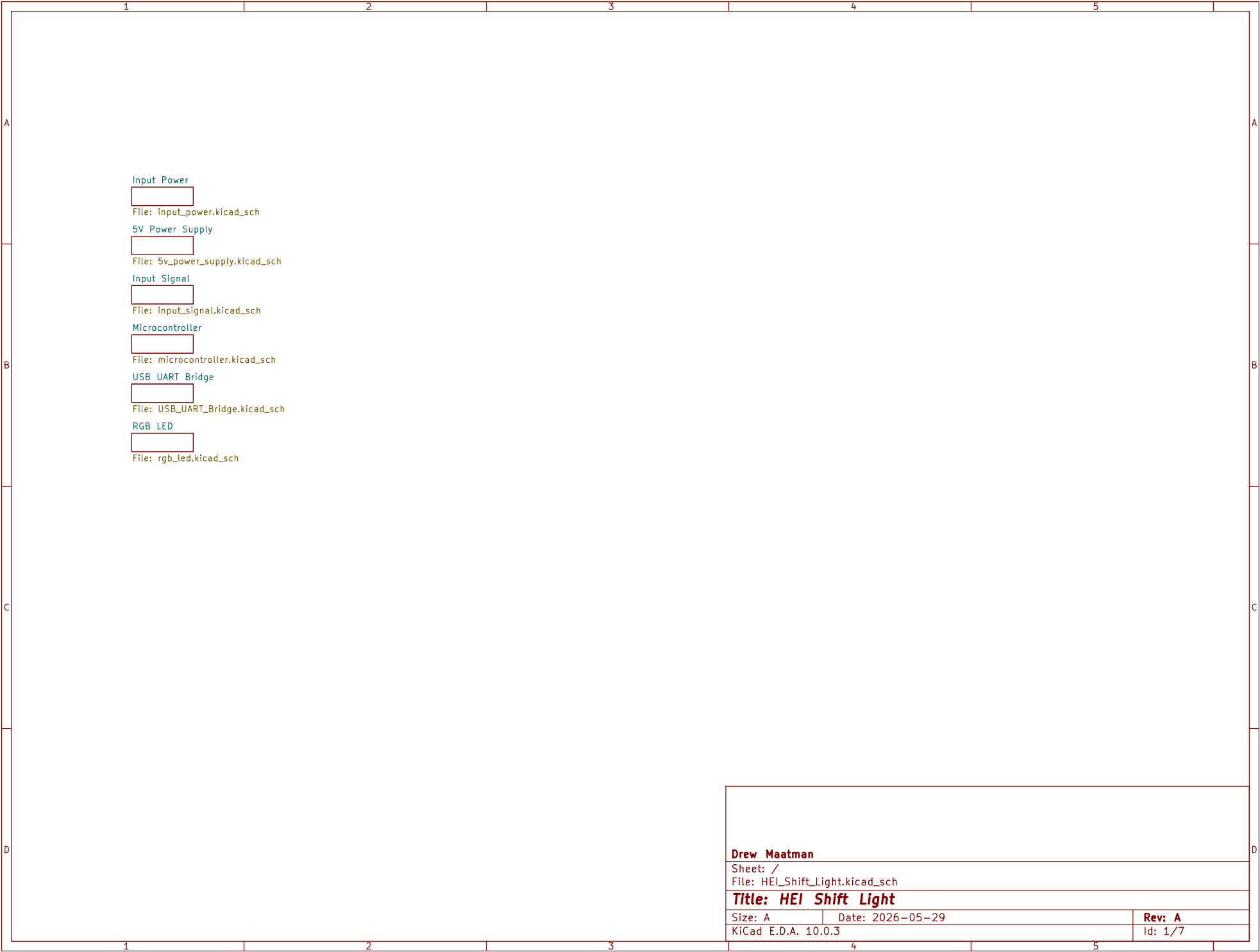
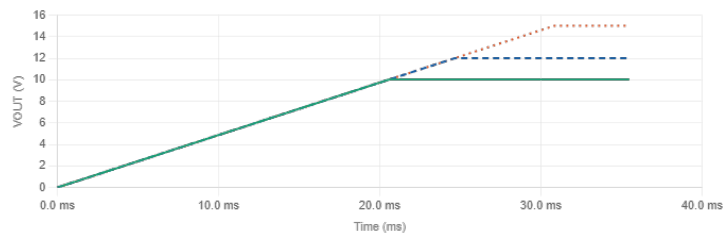
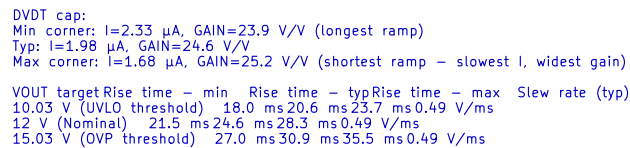






```

VIN_PG00D divider:
VEN at VUV (min VOUT)
2.98 V
Must be  $\geq 0.65$  V ☐
VEN at VOV (max VOUT)
4.46 V
☐ within safe range
Divider current at VUV
70.53  $\mu$ A
Keep  $\geq 10\times$  EN leakage
V(EN) min = 2.98 V (need  $\geq 0.65$  V) ☐ V(EN) max = 4.46 V
Fault: FLT pulls EN to -0 V (< 0.3 V)

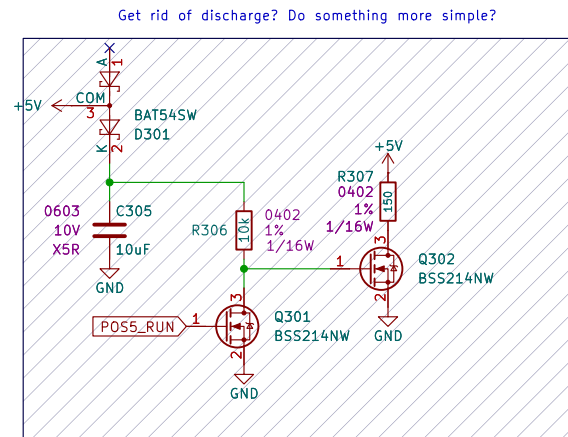
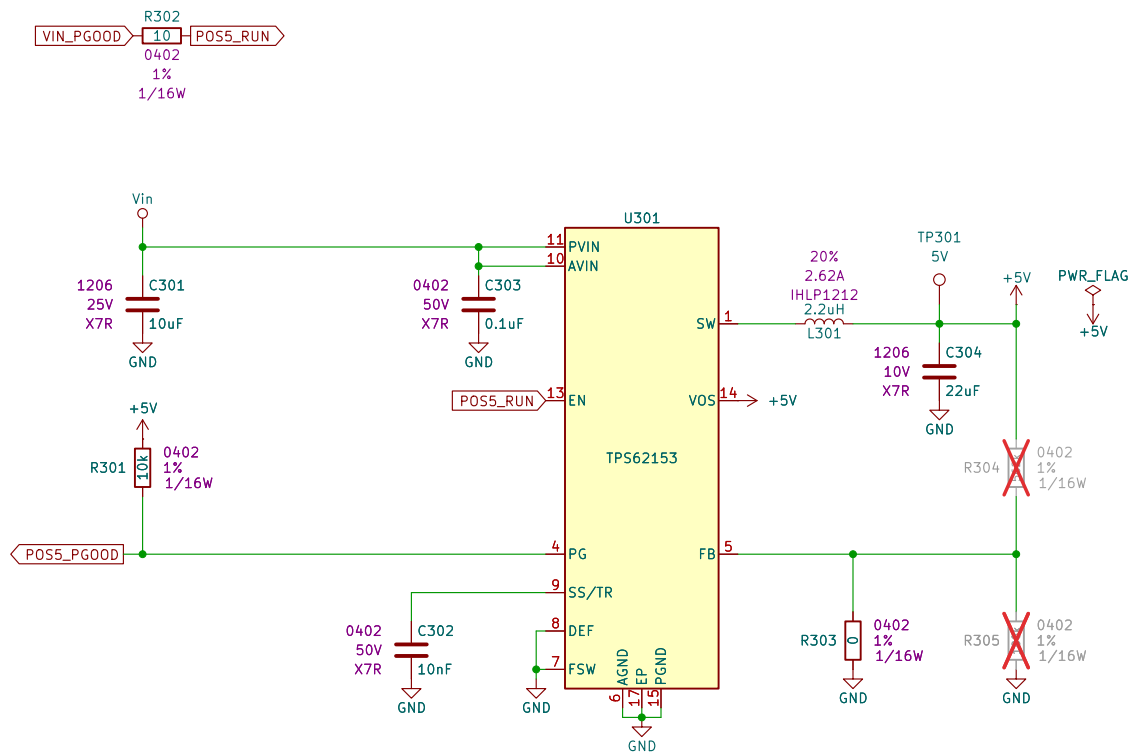
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Sheet: /Input Power/
File: input_power.kicad_sch

Title: HEI Shift Light

Size: A	Date: 2026-05-29
KiCad E.D.A. 10.0.3	

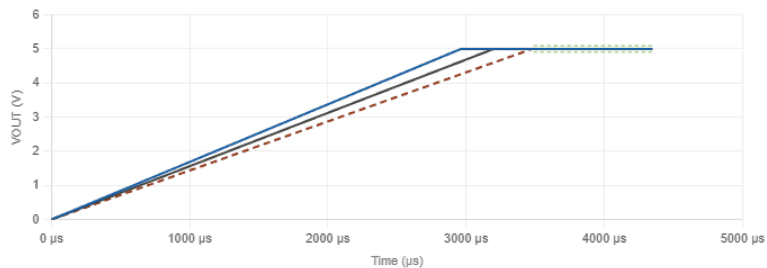
Rev: A
Id: 2/7



Rise times & settled output range

Min time: I=2.7 μ A (fastest) | Typ: I=2.5 μ A | Max time: I=2.3 μ A (slowest)

Condition Rise time - min Rise time - typ Rise time - max Settled VOUT
 DEF=low (nominal 5.00 V) 2.963 ms 3.200 ms 3.478 ms 4.910 - 5.090 V
 DEF=high (+5% = 5.25 V) 2.963 ms 3.200 ms 3.478 ms 5.155 - 5.345 V



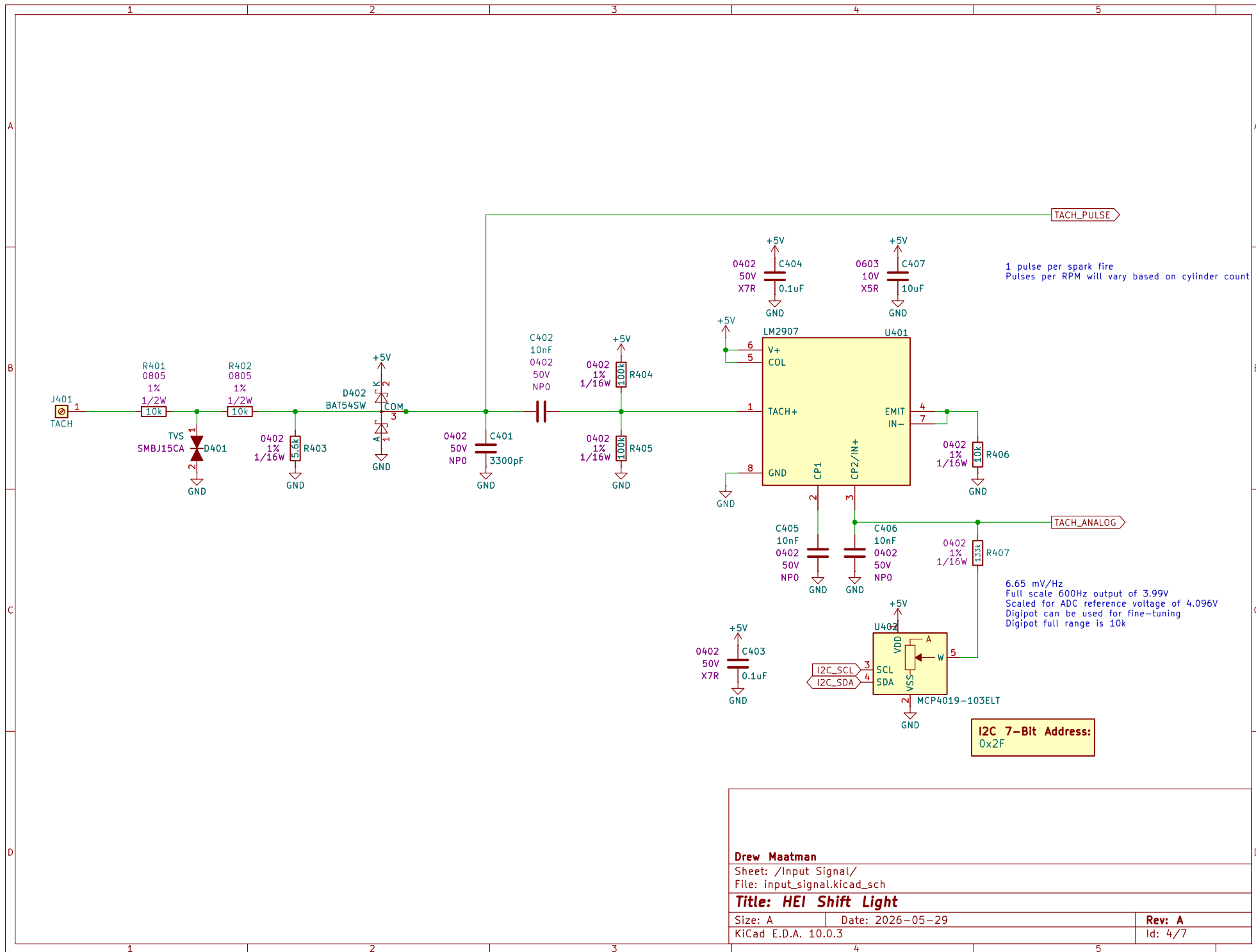
Drew Maatman

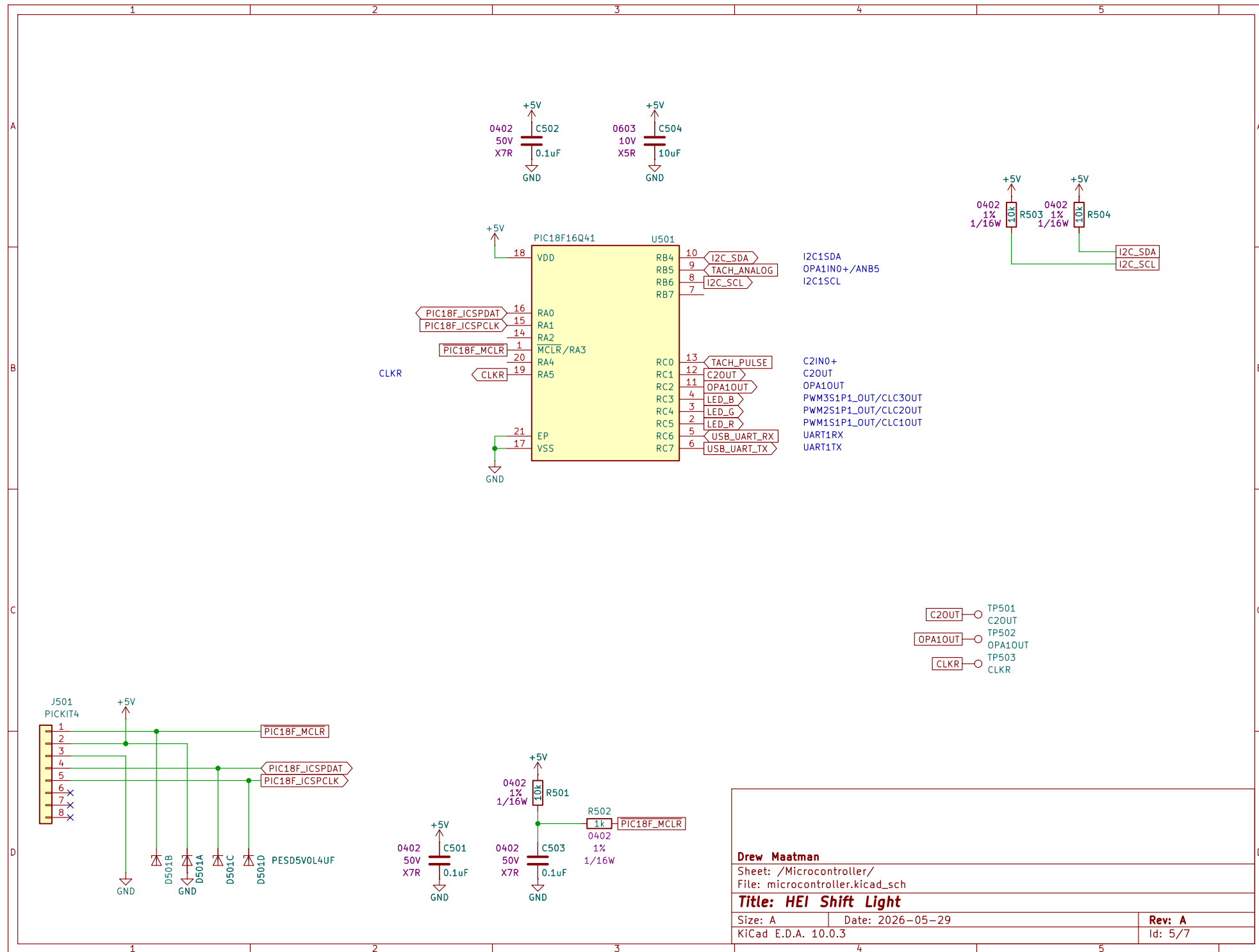
Sheet: /5V Power Supply/
 File: 5v_power_supply.kicad_sch

Title: HEI Shift Light

Size: A Date: 2026-05-29
 KiCad E.D.A. 10.0.3

Rev: A
 Id: 3/7





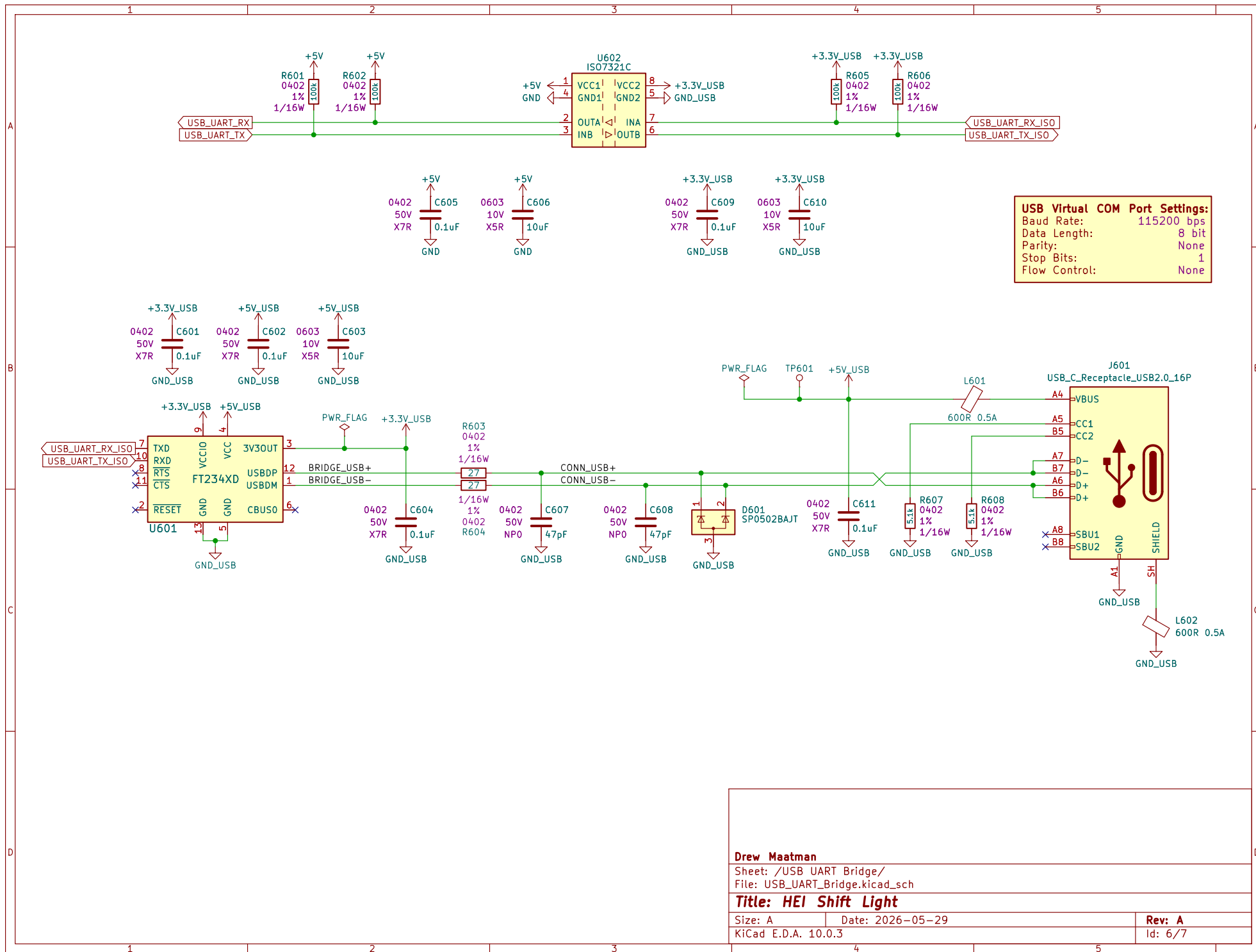
Drew Maatman

Sheet: /Microcontroller/
File: microcontroller.kicad_sch

Title: HEI Shift Light

Size: A Date: 2026-05-29
KiCad E.D.A. 10.0.3

Rev: A
Id: 5/7



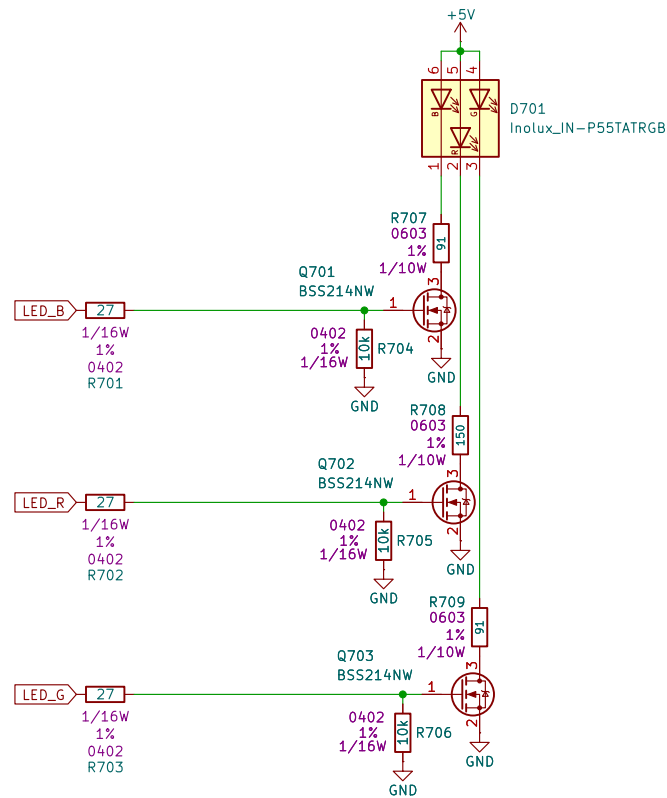
Drew Maatman

Sheet: /USB UART Bridge/
File: USB_UART_Bridge.kicad_sch

Title: HEI Shift Light

Size: A Date: 2026-05-29
KiCad E.D.A. 10.0.3

Rev: A
Id: 6/7



Based on the datasheet, the typical forward current (I_F) for testing is 20mA, with an absolute maximum of 25mA.
For "maximum brightness" while maintaining reliability, I recommend targeting 20mA.

Using Ohm's Law ($R = (V_{CC} - V_F) / I_F$ with $V_{CC} = 5V$:

Channel	Typ V_f	Calculation (20mA)	Recommended Res.	Max Bright (25mA)
Red	2.0V	$(5 - 2.0) / 0.020$	150 Ohm	120 Ohm
Green	3.2V	$(5 - 3.2) / 0.020$	91 Ohm (Standard)	72-75 Ohm
Blue	3.2V	$(5 - 3.2) / 0.020$	91 Ohm (Standard)	72-75 Ohm

Design Notes:

- Power Dissipation: At 20mA, the resistors dissipate approx 36-60mW. Standard 0603 or 0805 SMD resistors are more than sufficient.
- Safety: I strongly recommend the 20mA values. Pushing to 25mA provides a marginal increase in brightness but reduces the LED's lifespan and increases heat.

Drew Maatman

Sheet: /RGB LED/
File: rgb_led.kicad_sch

Title: HEI Shift Light

Size: A Date: 2026-05-29
KiCad E.D.A. 10.0.3

Rev: A
Id: 7/7